

HETEROGENEOUS SOC RESOURCE CONTENTION & POWER-RAIL INTERFERENCE

Shared Memory Crossbar Burst Starvation and Transient PDN Voltage Droop on Shared Silicon

(a) Memory Bus & Cache Contention

64-bit LPDDR4 · 12.8 GB/s

PROPOSAL ENGINE (NPU / MPU)

- 50 Hz Policy Inference
- 180 MB weights & activations
- **256 kB burst @ 12.0 GB/s (94% bus)**

SAFETY ENFORCER (RT Core)

- 1000 Hz Hard Real-Time Loop
- 64 kB sensor history & invariants
- **Untended read budget: 5.0 μ s**

SHARED SYSTEM CROSSBAR (AXI4/5 QoS) & 8MB LAST-LEVEL CACHE

- ⚡ Matrix tile burst evicts enforcer sensor tables from L3 ($t_{\text{refill}} = 20\text{ns} \rightarrow 350\text{ns}$)
- Non-preemptive packet-atomic bursts ($\text{ARLEN} \leq 256$ beats cannot be truncated)

DRAM CONTROLLER QUEUE & ARBITER (Head-of-Line Blocking)

In-Flight NPU Burst

256 kB Tile (20.0 μ s)

Sensor DMA Ingest

64 kB (5.0 μ s)

Log Dump

32 kB (2.5 μ s)

RT Enforcer

WAITS in Q

Total Queuing Delay: $t_{\text{wait}} = 20.0 + 5.0 + 2.5 + 0.1$ (bank conflict) = 27.6 μ s

DRAM Page Reordering: Arbiter favors open row locality, starving real-time closed-bank reads.

Result: Enforcer read latency explodes from nominal 5.0 μ s to 27.6–80.0 μ s under load.

ENFORCEMENT TIMING SLACK COLLAPSE (1.0 ms Control Cycle)

1. **Untended (Cold / Idle): Total = 100.0 μ s (Slack = +20.0 μ s ✓)** Max Compute Budget (120 μ s)



2. **Contended Under NPU Burst: Total = 127.6 μ s (Slack = -7.6 μ s ✗)**



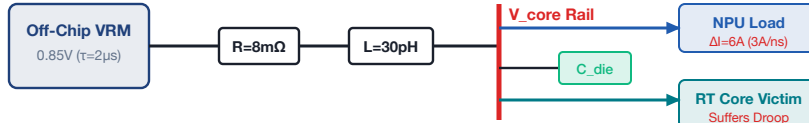
DEADLINE BREACH: 7.6 μ s Overrun into Valve Transit Margin

Although average channel utilization is only 25%, instantaneous burst collisions exhaust timing margins and delay high-pressure shutoff command.

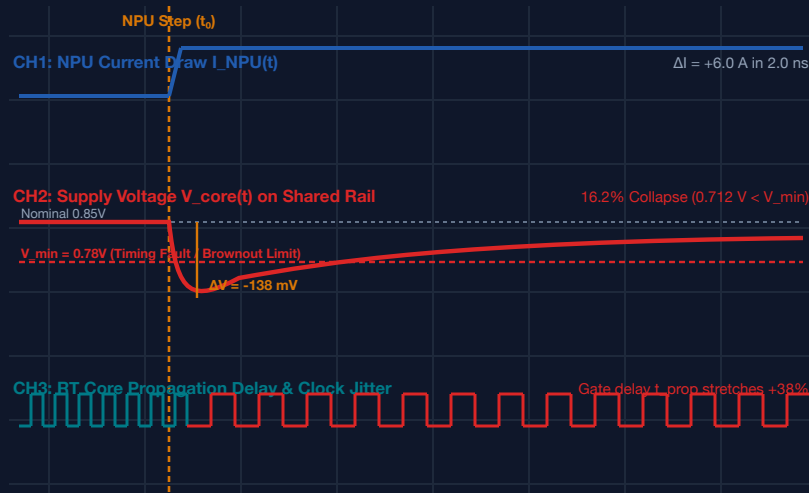
(b) PDN Transient Voltage Droop & Clock Jitter

$\Delta V = L(di/dt) + I \cdot R$

POWER DISTRIBUTION NETWORK (PDN) EQUIVALENT CIRCUIT



Droop calculation: $\Delta V = L(di/dt) + I \cdot R = 30\text{pH} \cdot 3\text{A/ns} + 6\text{A} \cdot 8\text{m}\Omega = 90\text{mV} + 48\text{mV} = 138\text{mV}$



PHYSICAL CROSS-DOMAIN FAILURE COUPLING

- VRM loop bandwidth (<1 MHz) is 3 orders of magnitude too slow for nanosecond di/dt steps.
- Logic propagation delay t_{prop} proportional to $V_{\text{DD}}/(V_{\text{DD}} - V_{\text{th}})^2$ causes setup-time violations.
- DVFS emergency throttle cuts clock to 600 MHz: $t_{\text{exec}} = 0.70\text{ms} \rightarrow 1.40\text{ms} > 1.0\text{ms}$ deadline.
- Verdict: Software isolation is defeated by power grid impedance and thermal substrate coupling.**